

Lecture 8: Text categorization

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Applications of text categorization

- Automatically classify political news from sports news

political

POLITICS | WHITE HOUSE MEMO

Gloom Lifts, and Obama Goes All Out

By MICHAEL D. SHEAR JAN. 21, 2015

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Obama's Zinger in State of Union Address
Video by Associated Press on January 20, 2015. Photo by Doug Mills/The New York Times.

sports

PRO FOOTBALL | ANALYSIS

Super Bowl 2015: Patriots' Red-Hot Offense Faces Seahawks' Dominant Defense

By CHASE STUART JAN. 20, 2015

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Last year's Super Bowl pitted one of the greatest single-season offenses in N.F.L. history against one of the greatest single-season defenses. Using slightly different time frames, this year's Super Bowl can boast similar claims.

Both the New England Patriots and the Seattle Seahawks had slow starts in 2014. After New England's 41-14 loss to the Kansas City Chiefs in Week 4, pundits wondered if we were witnessing the end of the Tom Brady/Bill Belichick-era Patriots. But since that game, the offensive line emerged as a cohesive unit, Rob Gronkowski's health improved and Brady became red-hot. Since that



The Seahawks' Richard Sherman intercepting a pass against the Packers in the N.F.C. title game.
David J. Phillip/Associated Press

Applications of text categorization

- Recognizing spam emails

```
Received: from 192.168.1.100 ([65.202.85.3]) by pacific-carrier-annex.mit.edu  
(8.9.2/8.9.2) with SMTP id AAA06179;  
Mon, 11 Jun 2001 00:39:32 -0400 (EDT)  
From: [some forged email address]  
Message-ID: <200106110439.AAA06179@pacific-carrier-annex.mit.edu>  
Subject: I am as shocked as you!  
Date: Sun, 10 Jun 01 00:32:35 Pacific Daylight Time  
X-Priority: 3  
X-MSMailPriority: Normal  
Importance: Normal  
MIME-Version: 1.0  
Content-Type: multipart/mixed;  
boundary="-----_NextPart_000_018C_01BD9940.715D52A0"
```

<HTML>

<BODY>

Some of the most beautiful women in the world bare it all for you. Denise Richard
s, Britney Spears, Jessica Simpson, and many more. <A HREF="http://216.130.166.1
88/index.html">CLICK HERE FOR NUDE CELEBS

</BODY></HTML>

Spam=**True**/False

Applications of text categorization

- Sentiment analysis



The best tablet, but not a necessary one., November 25, 2014

By [Andy, an Amazon Customer](#) (Fargo, ND) - [See all my reviews](#)

This review is from: **Apple iPad Air 2 MH0W2LL/A (16GB, Wi-Fi, Gold) NEWEST VERSION (Personal Computers)**

Short version: if you don't have a tablet yet, this is the one to get holiday 2014. If you already have a tablet that you're mostly happy with, whether an iPad or Android version, keep it.

I purchased the new iPad Air 2, in Gold, 16GB capacity about a week ago at Walmart, and I'd like to give a few impressions of the hardware and software here. I had particularly high hopes for this device, and have been waiting a long time to buy one; after holding a friend's brand new 64GB version, and being really impressed by how light the device seemed, I bought one for myself! :)

A little bit of background: My other experience with tablets involves a 2013 Nexus 7 that I use at least weekly; an Asus Transformer Pad, with a Tegra 3 1920x1080 screen, an Acer android tablet whose screen cracked 3 months after purchase; a Kindle Fire HD; I have also used both an iPad 2 and an iPad Mini (original) off and on, but never owned an iPad before. I use an iPhone 5.

The device is extremely light and thin. Its shocking, honestly - its far lighter than my chunky Kindle Fire HD 7. I bought it in gold (because why not live a little?) and it looks really nice. It feels like a premium device. The back is metal, which can be a little cold to the touch, but is smooth and easy to hold. It does get tedious holding it up while lying in bed, however. Probably this is due part to the small side bezels; my palm or thumb was nearly always bumping the screen.

The screen is gorgeous. Bright, easy to read, and I haven't noticed any reflections on it yet, which is fantastic. Honestly, its beautiful. And it shows off photographs really really well. I haven't used it to take any pictures, and probably won't, so I can't really comment on that aspect.

The software is good, but I was honestly expecting something noticeably better than iOS 8 on my iPhone, which just isn't the case. In fact, because of the animations, and the larger screen, it feels almost slower than my two year old iPhone.

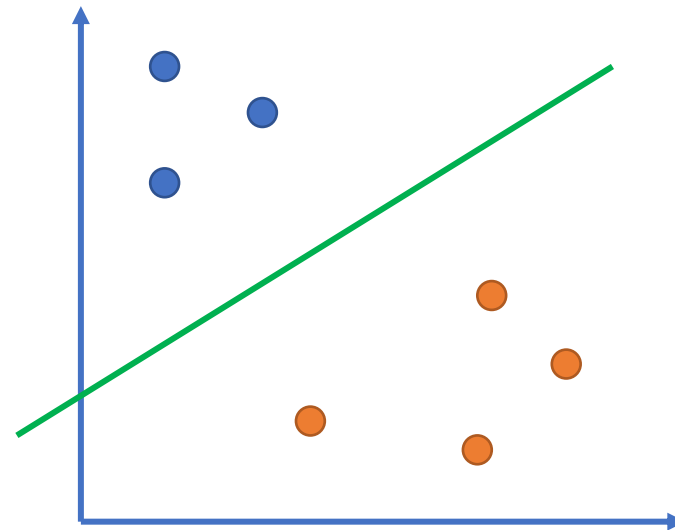
Basic notions about categorization

- Data points/Instances
 - X : an m -dimensional feature vector
- Labels
 - y : a categorical value from $\{1, \dots, k\}$
- Classification hyper-plane
 - $f(X) \rightarrow y$

vector space representation

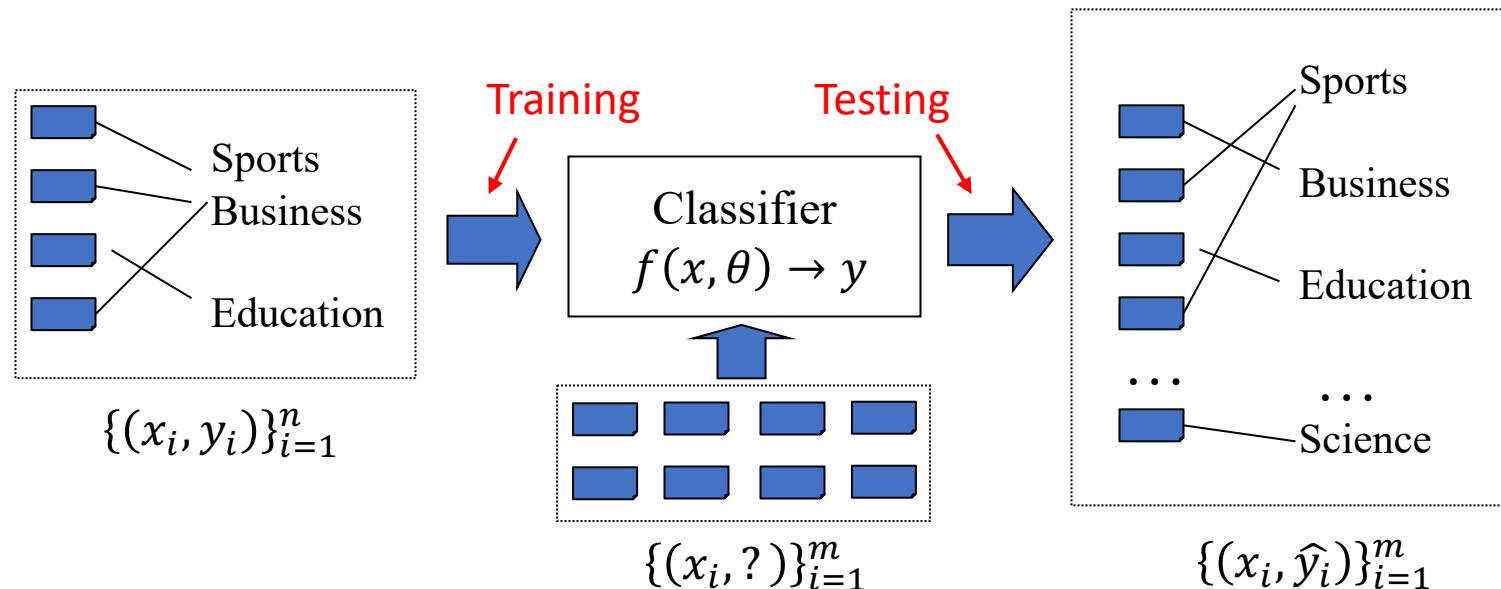


***Key question: how to
find such a mapping?***



Supervised text categorization

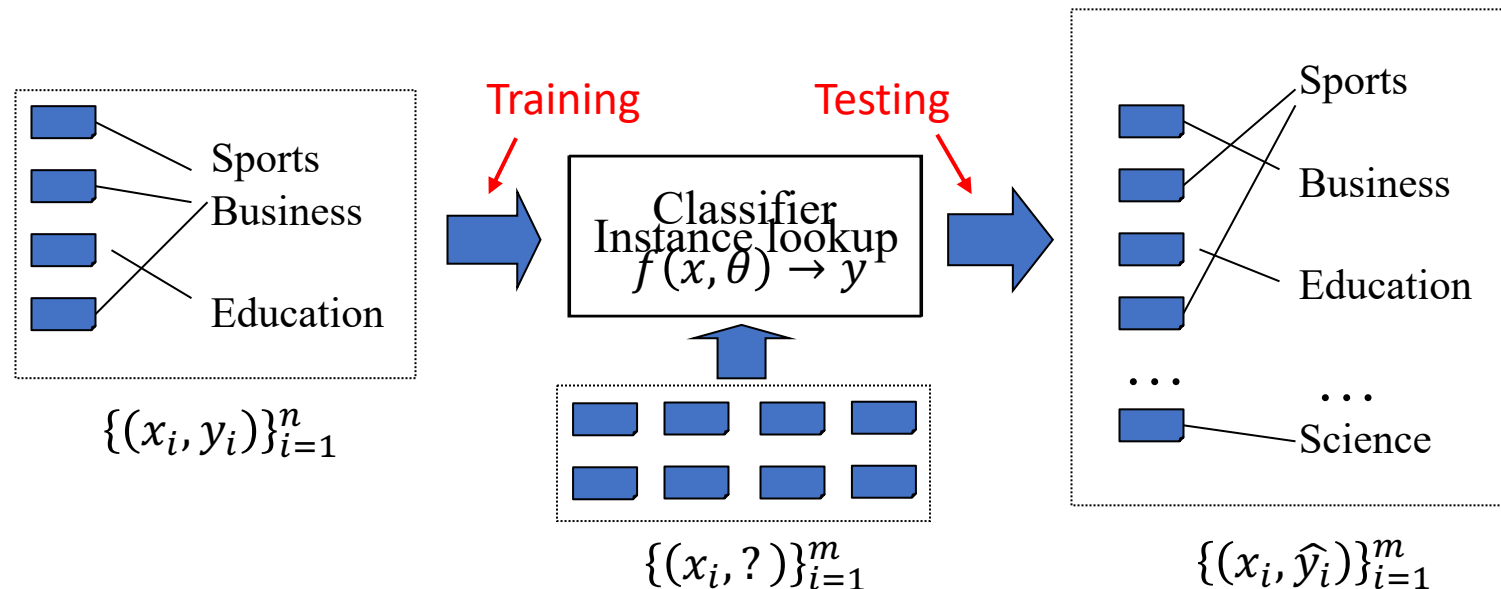
- Supervised learning
 - Estimate a model/method from labeled data
 - It can then be used to determine the labels of the unobserved samples



Type of classification methods

- Model-less
 - Instance based classifiers
 - Use observation directly
 - E.g., kNN

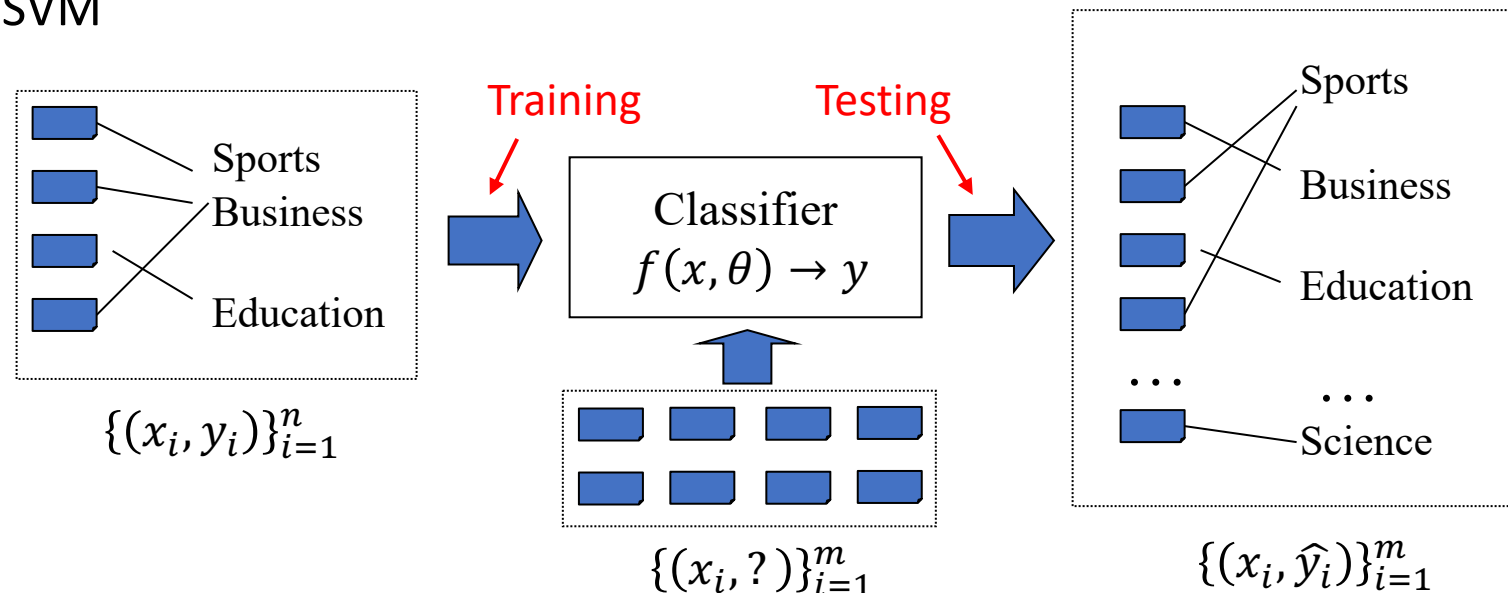
**Key: assuming similar items
have similar class labels!**



Type of classification methods

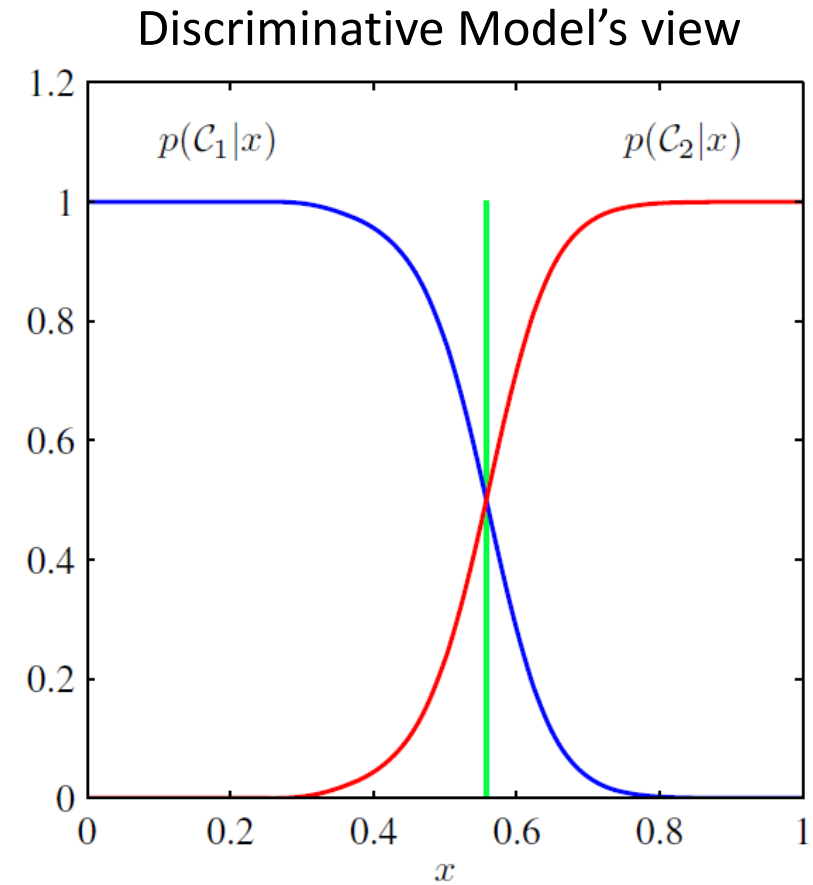
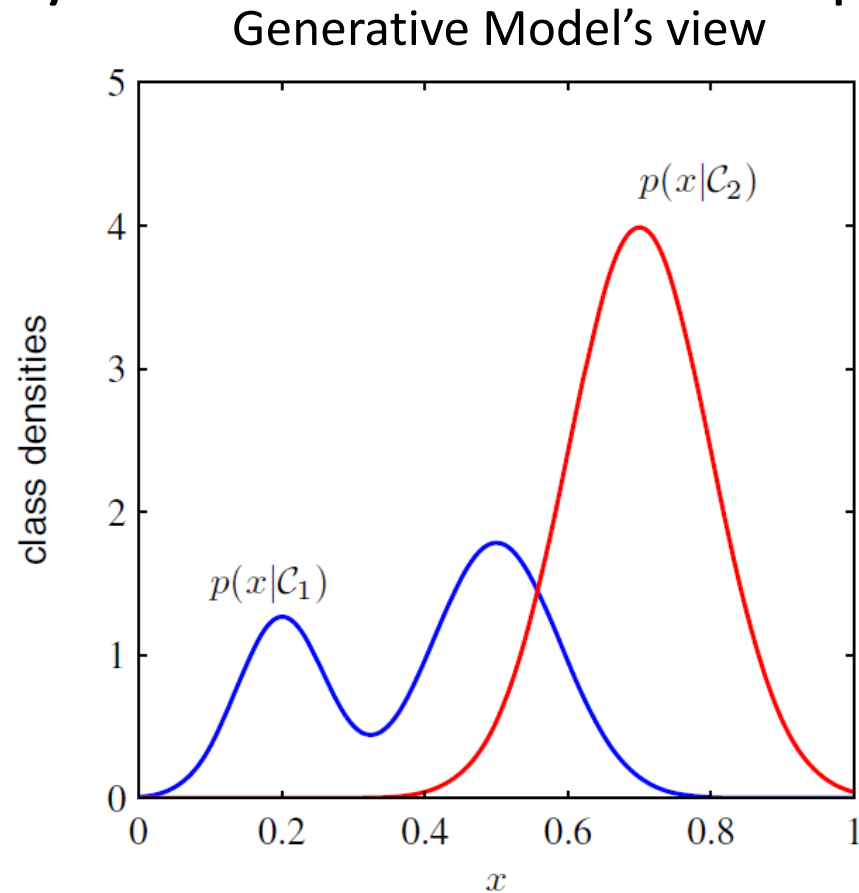
- Model-based
 - Generative models
 - Modeling joint probability of $p(x, y)$
 - E.g., Naïve Bayes
 - Discriminative models
 - Directly estimate a decision rule/boundary
 - E.g., SVM

Key: i.i.d. assumption!



Generative V.S. discriminative models

- Binary classification as an example



Generative V.S. discriminative models

Generative

- Specifying joint distribution
 - Full probabilistic specification for all the random variables
- Dependence assumption has to be specified for $p(x|y)$ and $p(y)$
- Flexible, can be used in unsupervised learning

Discriminative

- Specifying conditional distribution
 - Only explain the target variable
- Arbitrary features can be incorporated for modeling $p(y|x)$
- Need labeled data, only suitable for (semi-) supervised learning

General steps for text categorization

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Obama's Zinger in State of Union Address
Video by Associated Press on January 20, 2015. Photo by Doug Mills/The New York Times.



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1. Feature construction and selection
2. Model specification
3. Model estimation and selection
4. Evaluation

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1. Feature construction
and selection

2. Model specification

3. Model estimation and
selection

4. Evaluation

Consider:

1.1 How to represent the text documents?

1.2 Do we need all those features?

Feature construction for text categorization

- Vector space representation
 - Standard procedure in document representation
 - Features
 - N-gram, POS tags, named entities, topics
 - Feature value
 - Binary (presence/absence)
 - TF-IDF (many variants)

Feature selection for text categorization

- Select the most informative features for model training
 - Reduce noise in feature representation
 - Improve final classification performance
 - Improve training/testing efficiency
 - Less time complexity
 - Fewer training data

Feature selection methods

- Filter method
 - Evaluate the features independently from the classifier and other features
 - No indication of a classifier's performance on the selected features
 - No dependency among the features
 - Feasible for very large feature set
 - Usually used as a preprocessing step



R. Kohavi, G.H. John/Artificial Intelligence 97 (1997) 273-324

Feature scoring metrics

- Document frequency
 - Rare words: non-influential for global prediction, reduce vocabulary size

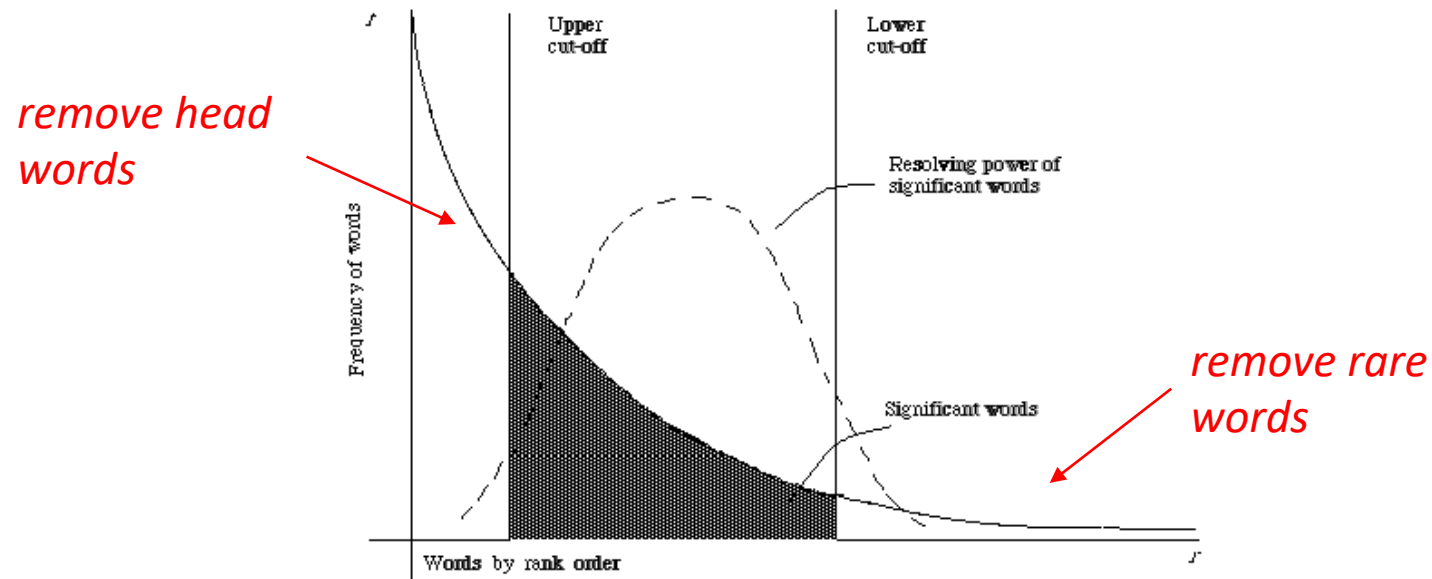


Figure 2.1. A plot of the hyperbolic curve relating f , the frequency of occurrence and r , the rank order (Adapted from Schultz⁴⁴ page 120)

Feature scoring metrics

- Information gain
 - Decrease in entropy of categorical prediction when the feature is present or absent

$$IG(t) = - \sum_c p(c) \log p(c) + p(t) \sum_c p(c|t) \log p(c|t) + p(\bar{t}) \sum_c p(c|\bar{t}) \log p(c|\bar{t})$$

Entropy of class label along

Entropy of class label if t is present

Entropy of class label if t is absent

probability of seeing class label c in documents where t occurs

probability of seeing class label c in documents where t does not occur

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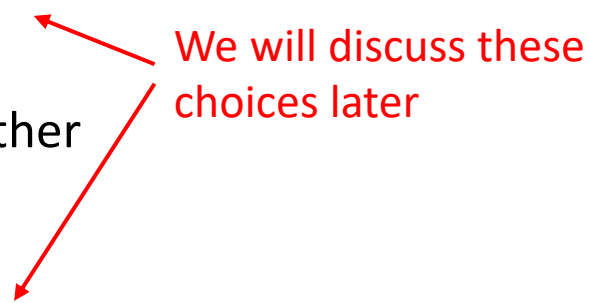
1. Feature construction and selection
2. Model specification
3. Model estimation and selection
4. Evaluation

Consider:

2.1 What is the unique property of this problem?

2.2 What type of classifier we should use?

Model specification

- Specify dependency assumptions
 - Linear relation between x and y
 - $w^T x \rightarrow y$
 - Features are independent among each other
 - Naïve Bayes, linear SVM
 - Non-linear relation between x and y
 - $f(x) \rightarrow y$, where $f(\cdot)$ is a non-linear function of x
 - Features are not independent among each other
 - Decision tree, kernel SVM, mixture model
 - Choose based on our domain knowledge of the problem
- 
- We will discuss these choices later

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Consider:

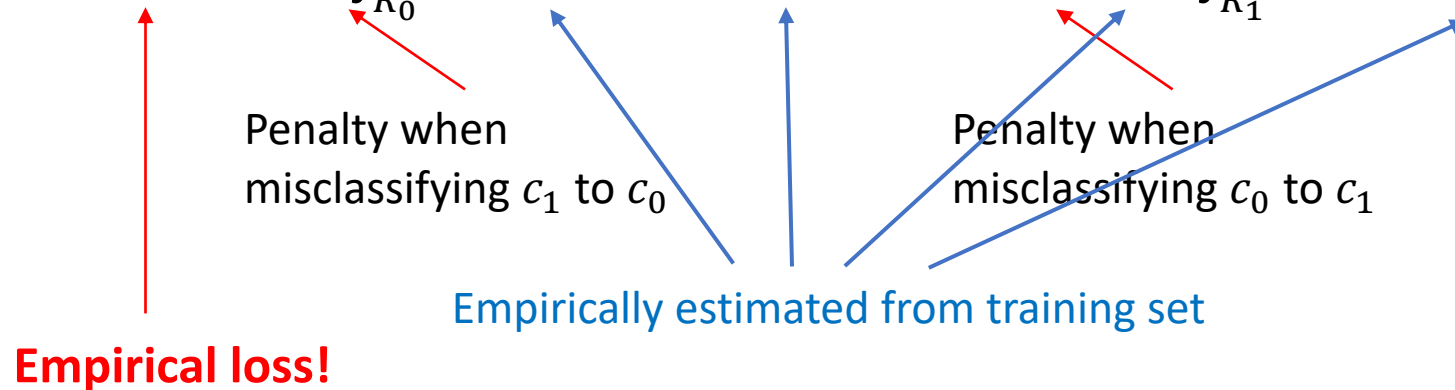
3.1 How to estimate the parameters in the selected model?

3.2 How to control the complexity of the estimated model?

Model estimation and selection

- General philosophy
 - Loss minimization

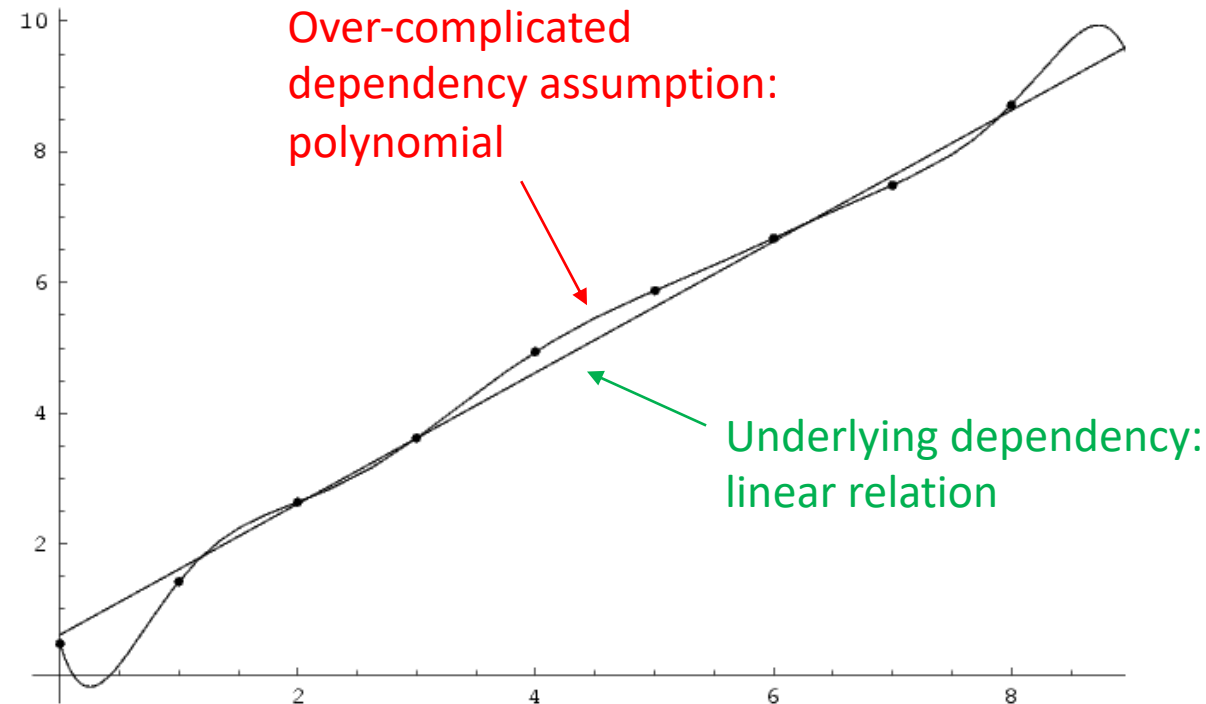
$$E[L] = L_{1,0}p(y = 1) \int_{R_0} p(x|y = 1)p(x)dx + L_{0,1}p(y = 0) \int_{R_1} p(x|y = 0)p(x)dx$$



Key assumption: Independent and Identically Distributed!

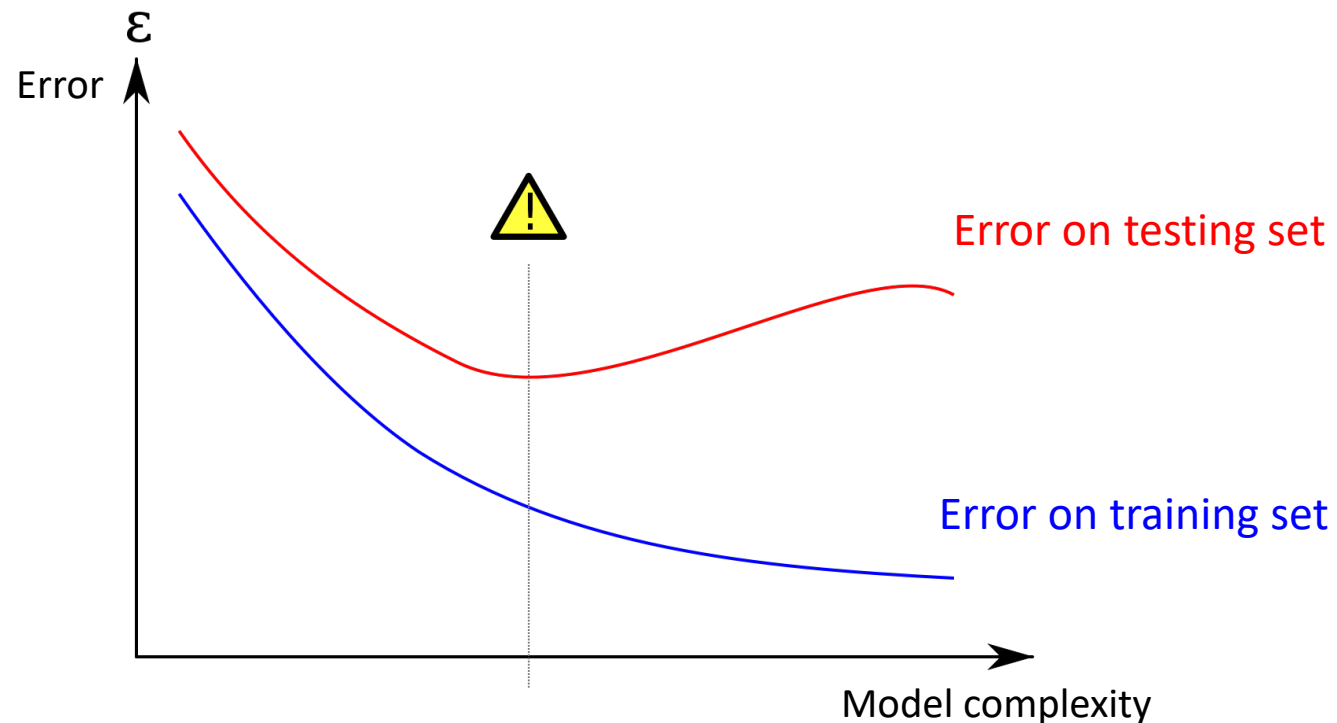
Empirical loss minimization

- Overfitting
 - Good empirical loss, terrible generalization loss
 - High model complexity -> prune to overfit noise



Generalization loss minimization

- Avoid overfitting
 - Measure model complexity as well
 - Model selection and regularization

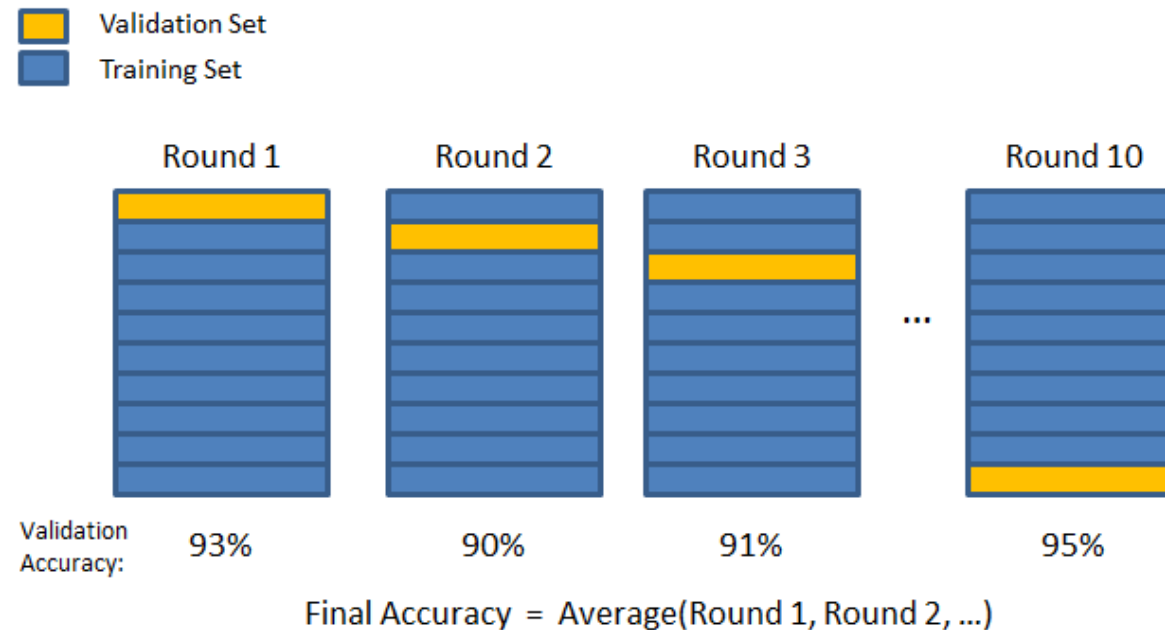


Generalization loss minimization

- Cross validation
 - Avoid noise in train/test separation
 - k -fold cross-validation
 1. Partition all training data into k equal size disjoint subsets;
 2. Leave one subset for validation and the other $k-1$ for training;
 3. Repeat step (2) k times with each of the k subsets used exactly once as the validation data.

Generalization loss minimization

- Cross validation
 - Avoid noise in train/test separation
 - k -fold cross-validation



Generalization loss minimization

- Cross validation
 - Avoid noise in train/test separation
 - k -fold cross-validation
 - Choose the model (among different models or same model with different settings) that has the best average performance on the validation sets
 - Some statistical test is needed to decide if one model is significantly better than another



Will cover it shortly

General steps for text categorization

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Consider:

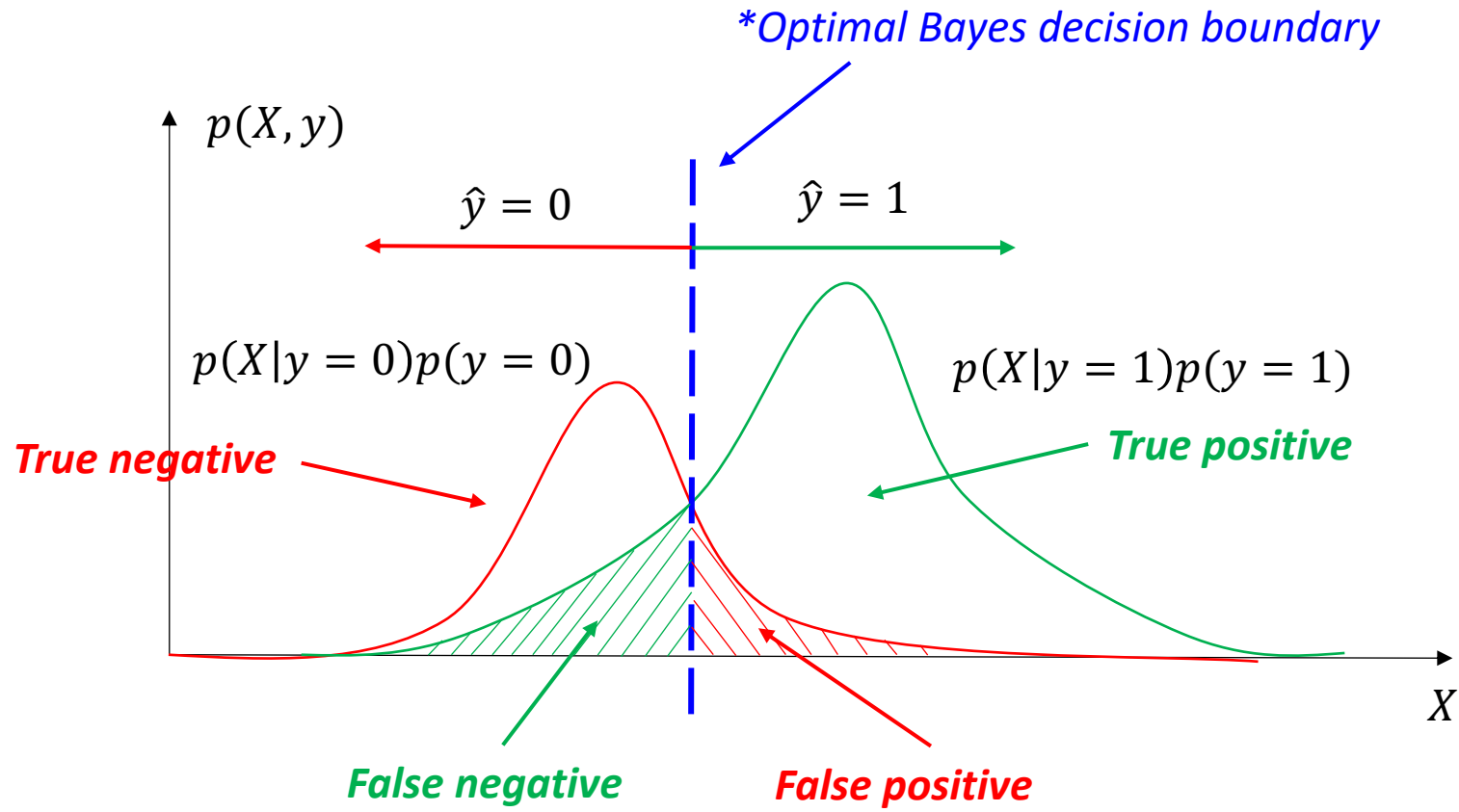
4.1 How to judge the quality of learned model?

4.2 How can you further improve the performance?

Classification evaluation

- Accuracy
 - Percentage of correct prediction over all predictions, i.e., $p(y^* = y)$
- Limitation
 - Highly skewed class distribution
 - $p(y^* = 1) = 0.99$
 - Trivial solution: all testing cases are positive
 - Classifiers' capability is only differentiated by 1% testing cases

Evaluation of binary classification



Summarizing precision and recall

- With a single value
 - In order to compare different classifiers
 - F-measure: weighted harmonic mean of precision and recall, α balances the trade-off

$$F = \frac{1}{\alpha \frac{1}{P} + (1 - \alpha) \frac{1}{R}} \quad \left(F_1 = \frac{2}{\frac{1}{P} + \frac{1}{R}} \right)$$

- Why harmonic mean?

- Classifier1: P:0.53, R:0.36
- Classifier2: P:0.01, R:0.99

*Equal weight between
precision and recall*

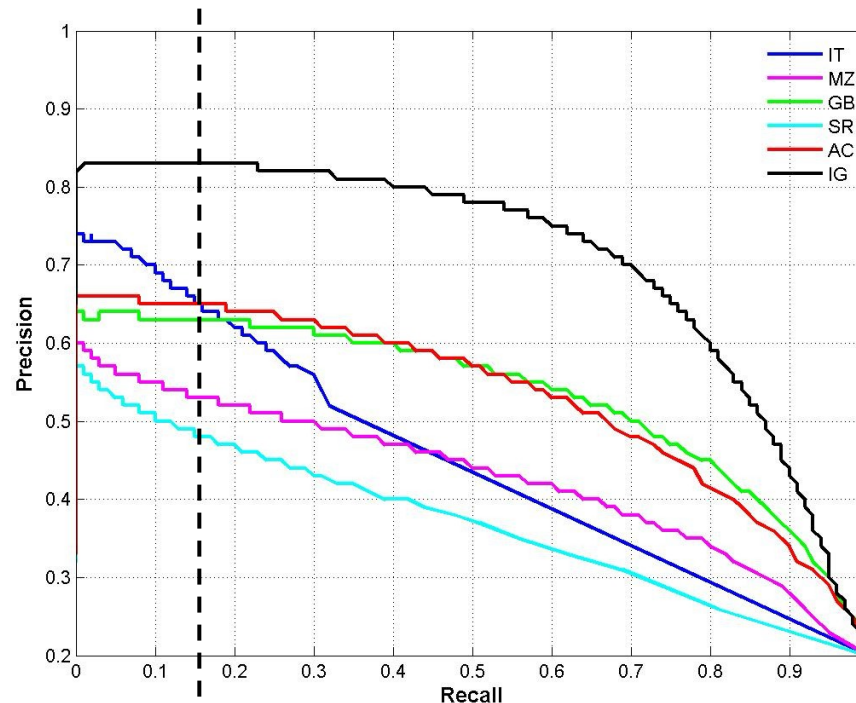
H	A
0.429	0.445
0.019	0.500

Summarizing precision and recall

- With a curve
 1. Order all the testing cases by the classifier's prediction score (assuming the higher the score is, the more likely it is positive);
 2. Scan through each testing case: treat all cases above it as positive (including itself), below it as negative; compute precision and recall;
 3. Plot precision and recall computed for each testing case in step (2).

Summarizing precision and recall

- With a curve
 - A.k.a., precision-recall curve
 - Area Under Curve (AUC)



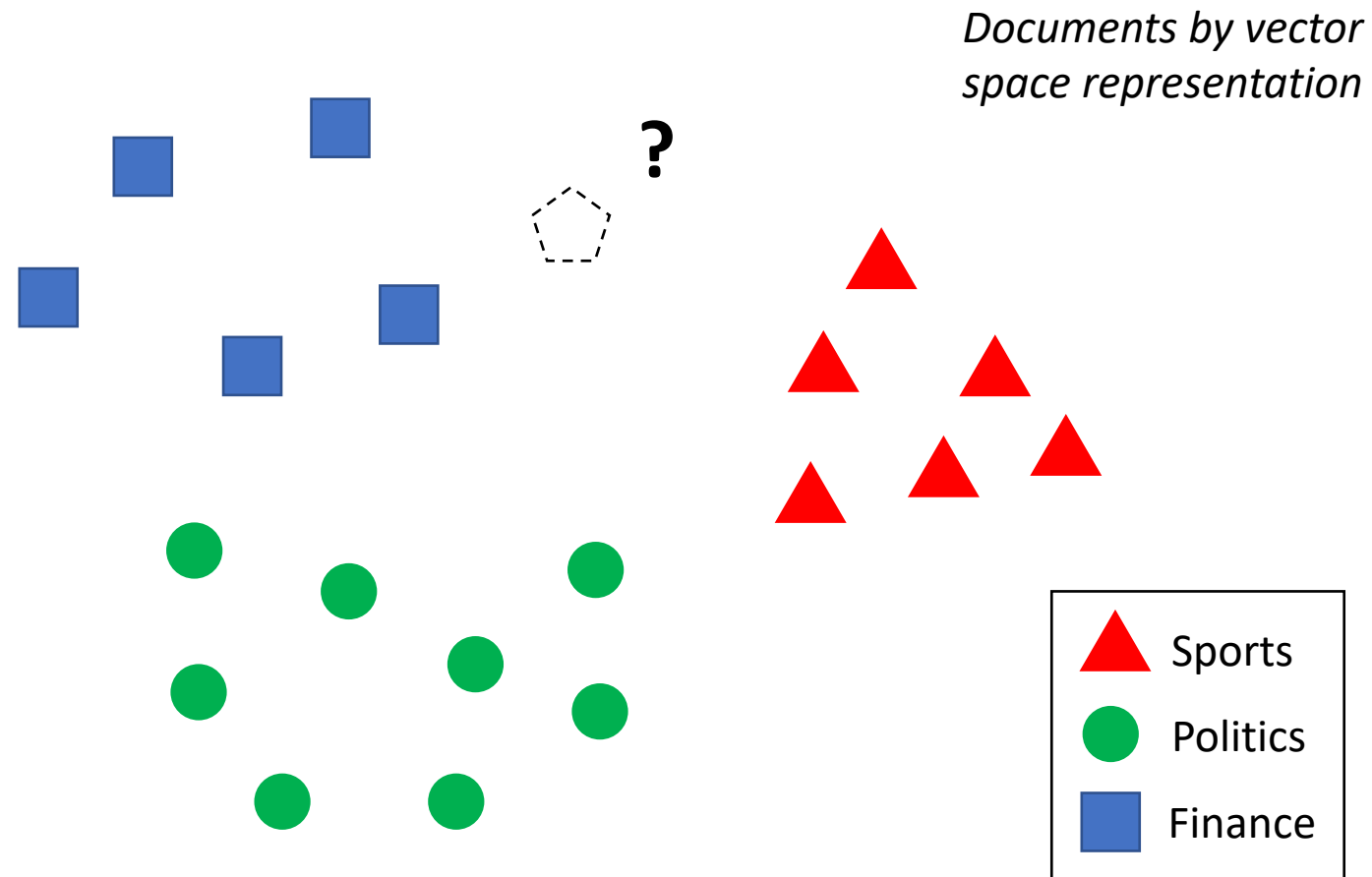
Under each recall level, we prefer a higher precision

Statistical significance tests

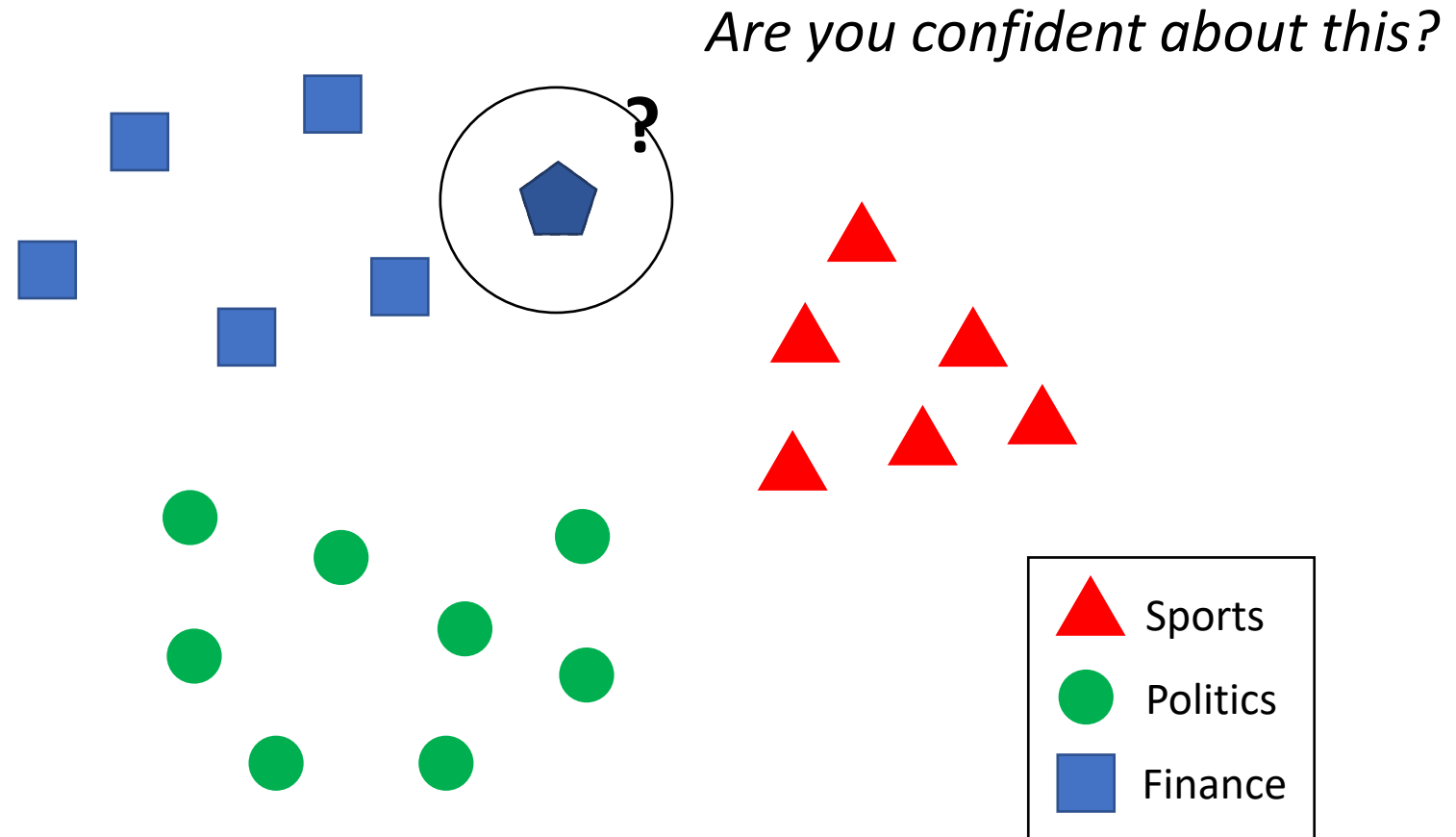
- How confident you are that an observed difference doesn't simply result from the train/test separation you chose?

<u>Fold</u>	<u>Classifier 1</u>	<u>Classifier 2</u>
1	0.20	0.18
2	0.21	0.19
3	0.22	0.21
4	0.19	0.20
5	0.18	0.37
Average	0.20	0.23

How to classify this document?

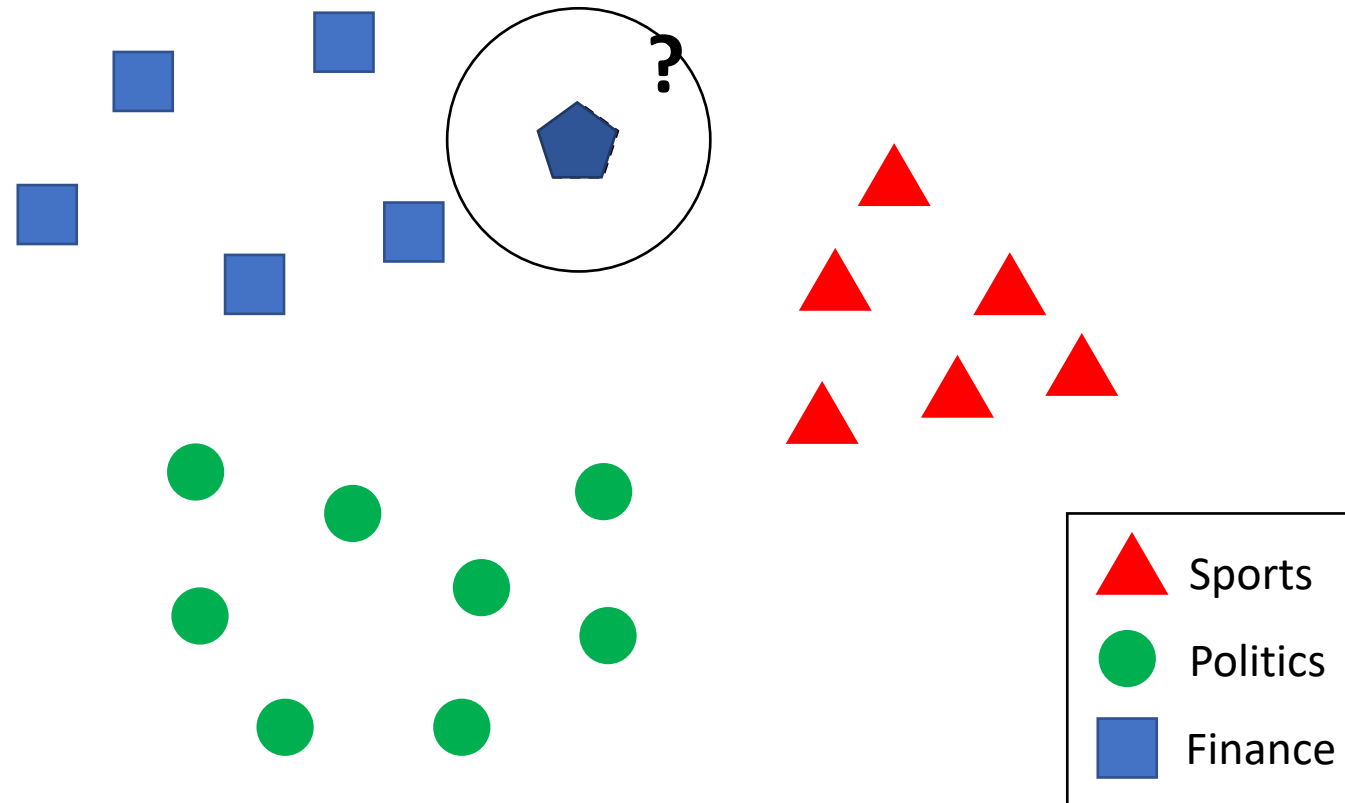


Let's check the nearest neighbor



Let's check more nearest neighbors

- Ask k nearest neighbors
 - Let them vote



Probabilistic interpretation of kNN

- Approximate Bayes decision rule in a subset of data around the testing point
- Let V be the volume of the m dimensional ball around x containing the k nearest neighbors for x , we have

$$p(x)V = \frac{k}{N} \Rightarrow p(x) = \frac{k}{NV}$$

Total number of instances

With Bayes rule:

$$p(y = 1|x) = \frac{\frac{N_1}{N} \times \frac{k_1}{N_1 V}}{\frac{k}{NV}} = \frac{k_1}{k}$$

Nearest neighbors from class 1

Total number of instances in class 1

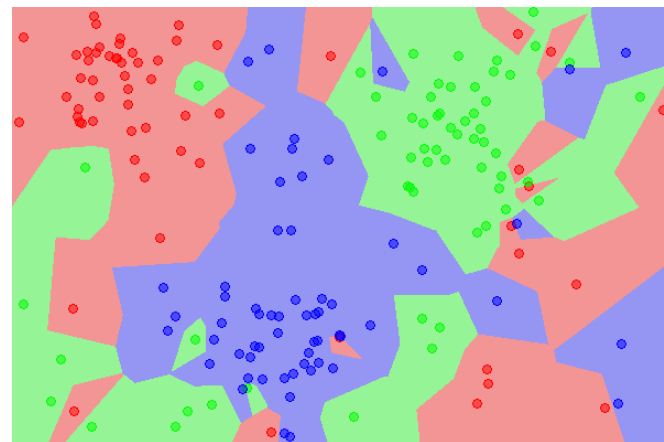
Counting the nearest neighbors from class 1

$p(x|y = 1) = \frac{k_1}{N_1 V}$

$p(y = 1) = \frac{N_1}{N}$

kNN is close to optimal

- Asymptotically, the error rate of 1-nearest-neighbor classification is less than twice of the Bayes error rate
- Decision boundary
 - 1NN - Voronoi tessellation



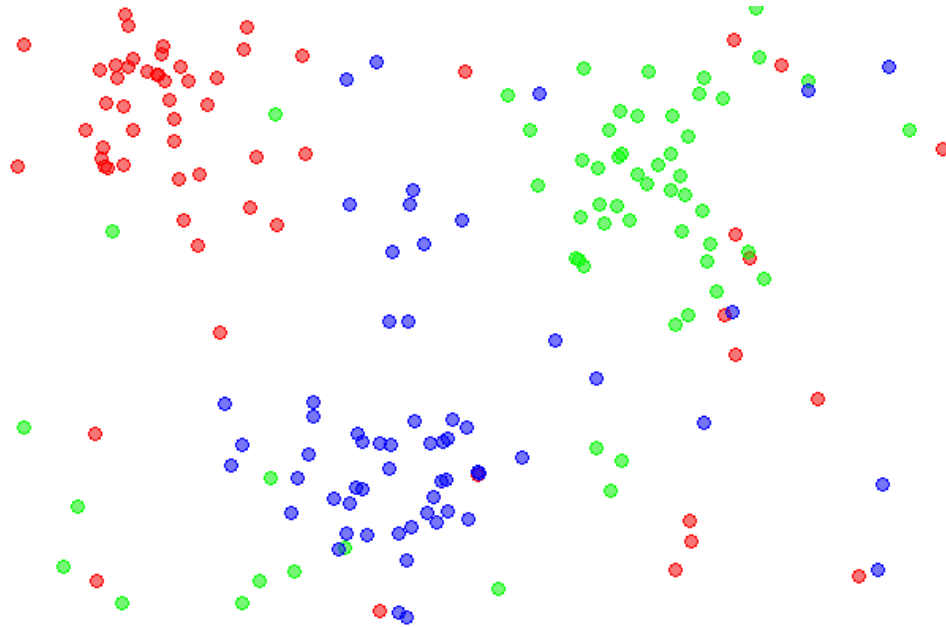
*A non-parametric estimation
of posterior distribution*

Components in kNN

- A distance metric
 - Euclidean distance/cosine similarity
- How many nearby neighbors to look at
 - k
- Instance look up
 - Efficiently search nearby points

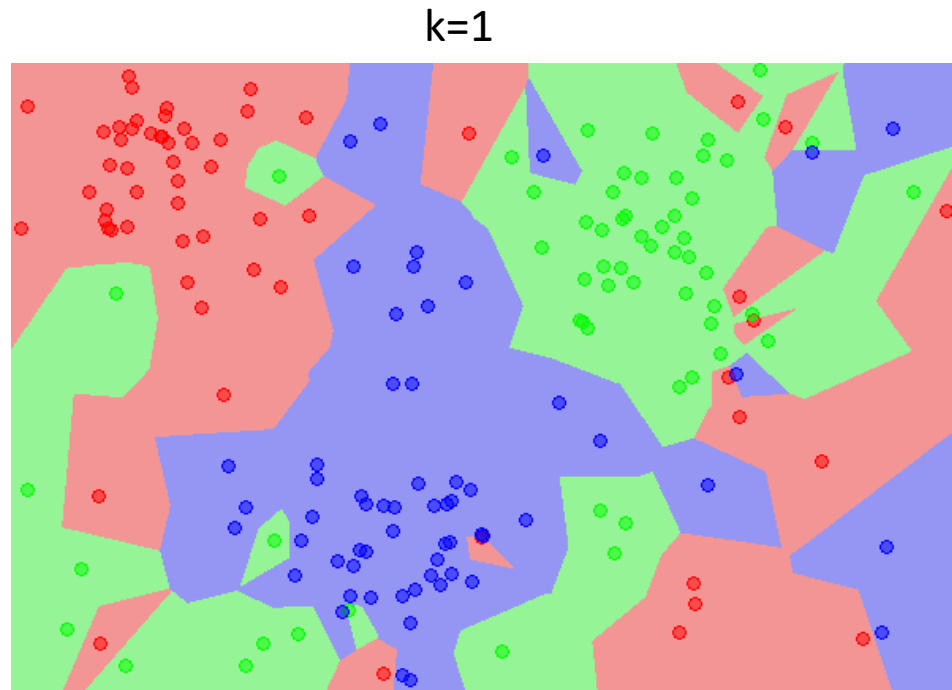
Effect of k

- Choice of k influences the “smoothness” of the resulting classifier



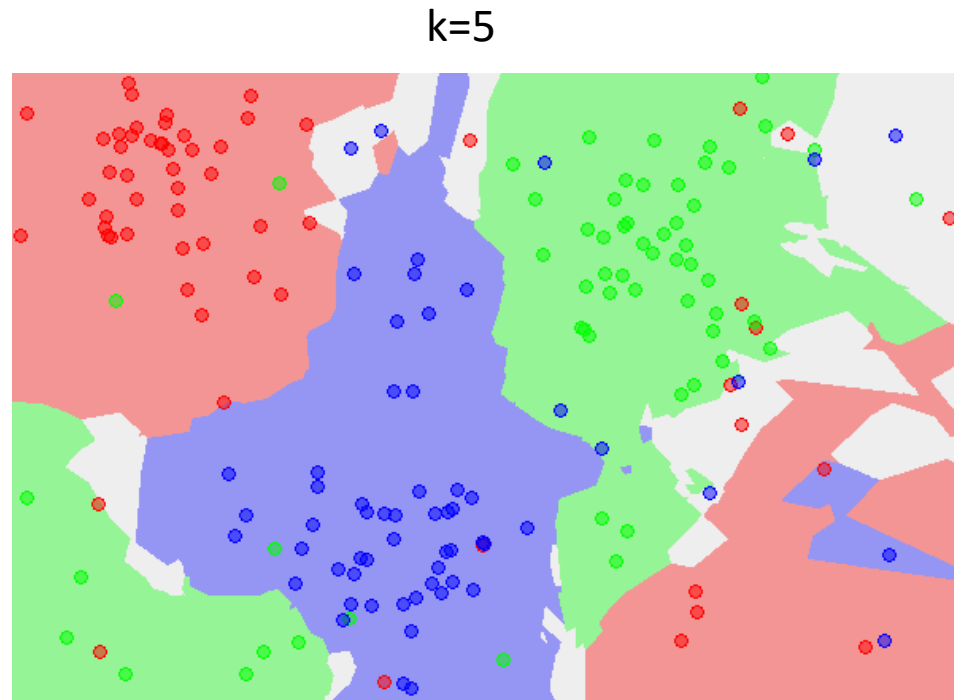
Effect of k

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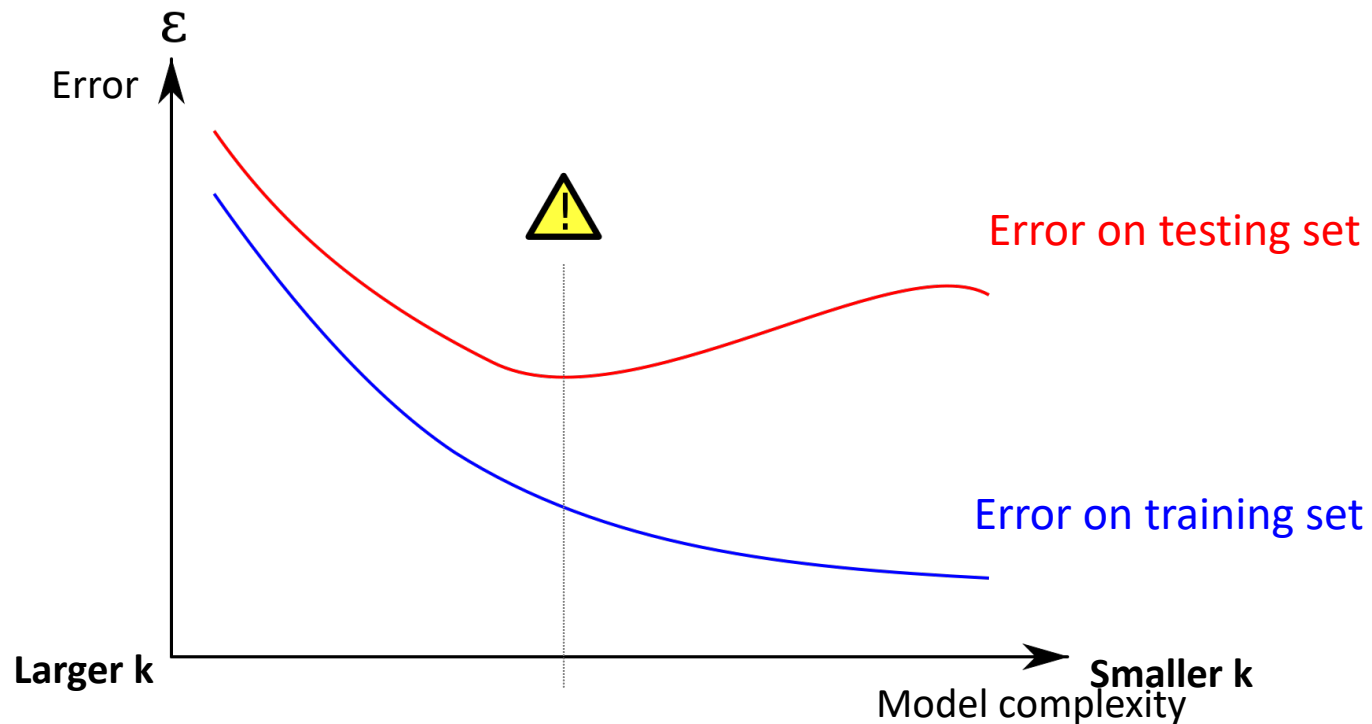
Effect of k

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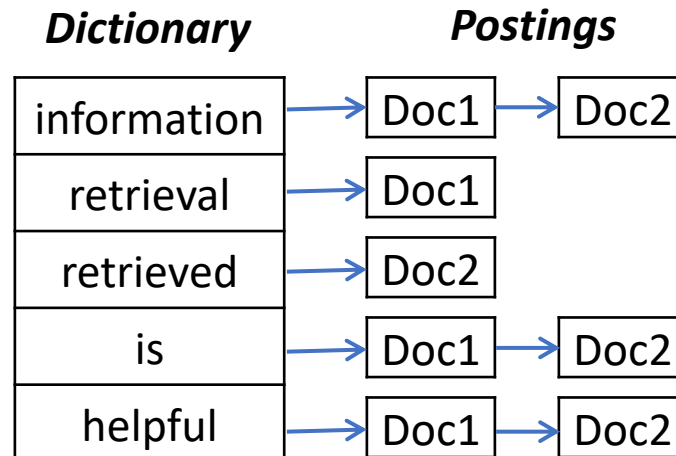
Effect of k

- Large k -> smooth shape for decision boundary
- Small k -> complicated decision boundary



Efficient instance look-up

- Exact solutions
 - Build inverted index for text documents
 - Special mapping: word -> document list
 - Speed-up is limited when average document length is large

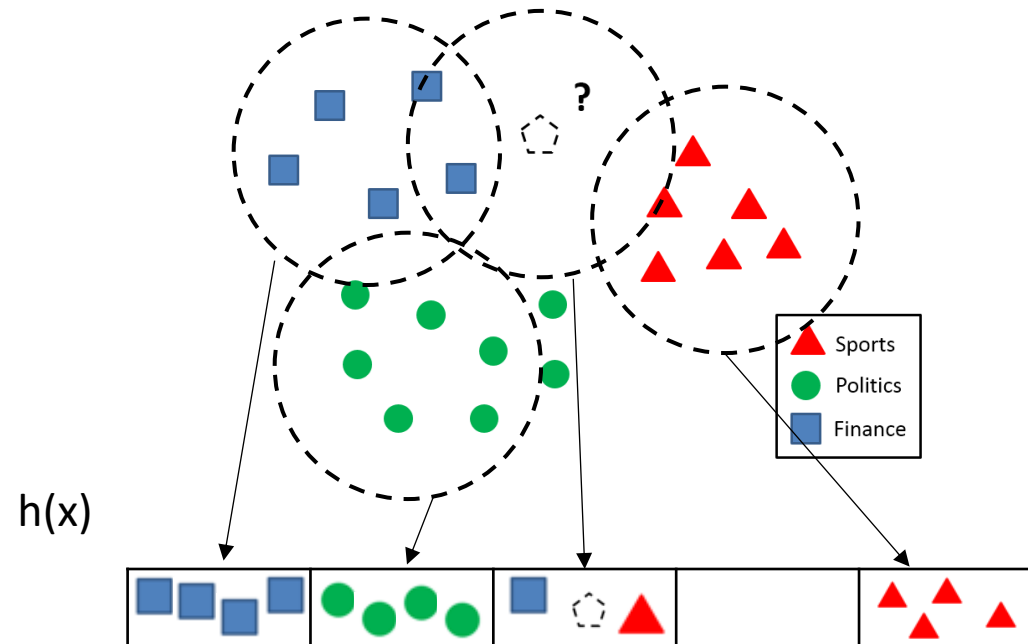


Efficient instance look-up

- Exact solutions
 - Build inverted index for text documents
 - Special mapping: word -> document list
 - Speed-up is limited when average document length is large
 - Parallelize the computation
 - Map-Reduce
 - Map training/testing data onto different reducers
 - Merge the nearest k neighbors from the reducers

Efficient instance look-up

- Approximate solution
 - Locality sensitive hashing
 - Similar documents -> (likely) same hash values



Efficient instance look-up

- Approximate solution
 - Locality sensitive hashing
 - Similar documents -> (likely) same hash values
 - Construct the hash function such that similar items map to the same “buckets” with a high probability
 - Learning-based: learn the hash function with annotated examples, e.g., must-link, cannot-link
 - Random projection

Summary

- Text categorization: basic idea
- Text classification

Readings

- [MRS] CH 13, CH 14.